

Game Analytics with Sportradar API: Technical Project Report

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Domain: Sports Analytics & Data Engineering

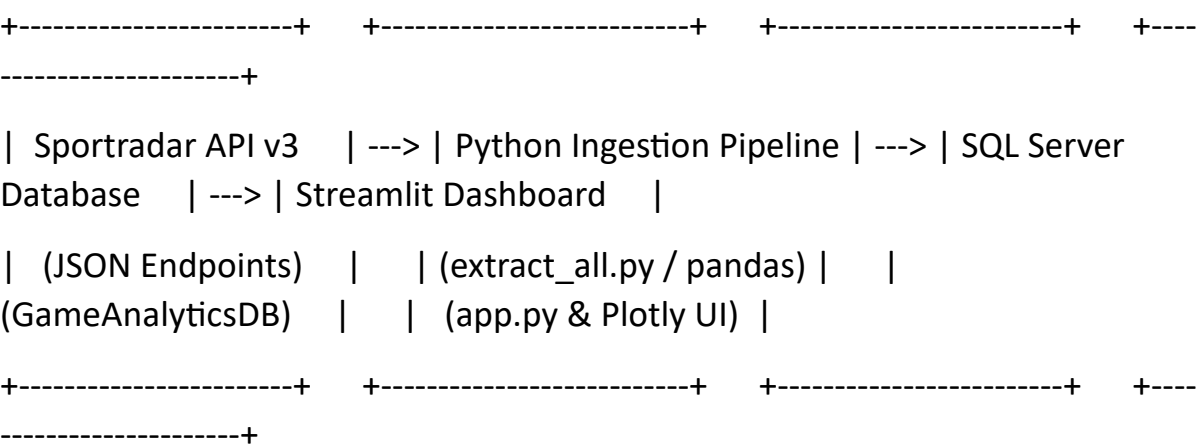
Tech Stack: Python 3, Microsoft SQL Server, Streamlit, Plotly, Pandas, SQLAlchemy

1. Executive Summary & Overview

The **SportRadar Event Explorer** is an end-to-end data engineering and interactive analytics platform designed to harvest, normalize, store, and visualize live tennis tournament and competitor data from the **Sportradar Tennis Trial API v3**.

The primary objective of this project is to automate the ingestion of semi-structured API payloads into a 3rd Normal Form (3NF) relational database hosted on Microsoft SQL Server, and to surface actionable sports insights through an interactive Streamlit web dashboard powered by Plotly.

2. End-to-End Workflow & Pipeline Architecture



Execution Stages

1. Extraction Layer (extract_all.py, fetch_complexes.py)

- Queries REST endpoints (Competitions, Complexes, Venues, Competitor Rankings).
- Implements custom protocol configurations (forcing IPv4 binding, non-persistent TCP socket headers) to navigate ISP Deep Packet Inspection (DPI) and rate-limiting constraints.

- Outputs raw JSON payloads into clean, intermediate CSV datasets.

2. Transformation Layer (pandas)

- Uses `pandas.json_normalize()` to flatten multi-nested JSON structures.
- Applies schema mappings, data type enforcement, null-value handling, and string truncations.

3. Storage Layer (load_to_sql.py)

- Programmatically establishes schemas and tables in Microsoft SQL Server (GameAnalyticsDB) via SQLAlchemy and pyodbc.
- Enforces strict Foreign Key sequencing to preserve referential integrity during bulk loads.

4. Presentation Layer (app.py)

- Hosts a Streamlit interactive UI connected to SQL Server.
- Features dynamic visual charts (Plotly), real-time KPI metrics, data filtering, and direct execution of 20 predefined analytical SQL queries.

3. Database Schema Design (3NF)

The relational database architecture is normalized to 3rd Normal Form (3NF) to eliminate data redundancy and preserve relational integrity across complex tournament structures.

+-----+		+-----+		+-----+
Categories	<-----	Competitions		Complexes
+-----+	1	* +-----+		+-----+
PK category_id		PK competition_id		1
category_name		competition_name		
+-----+		type		
		gender		*
		FK category_id		+-----+
+-----+		Venues		

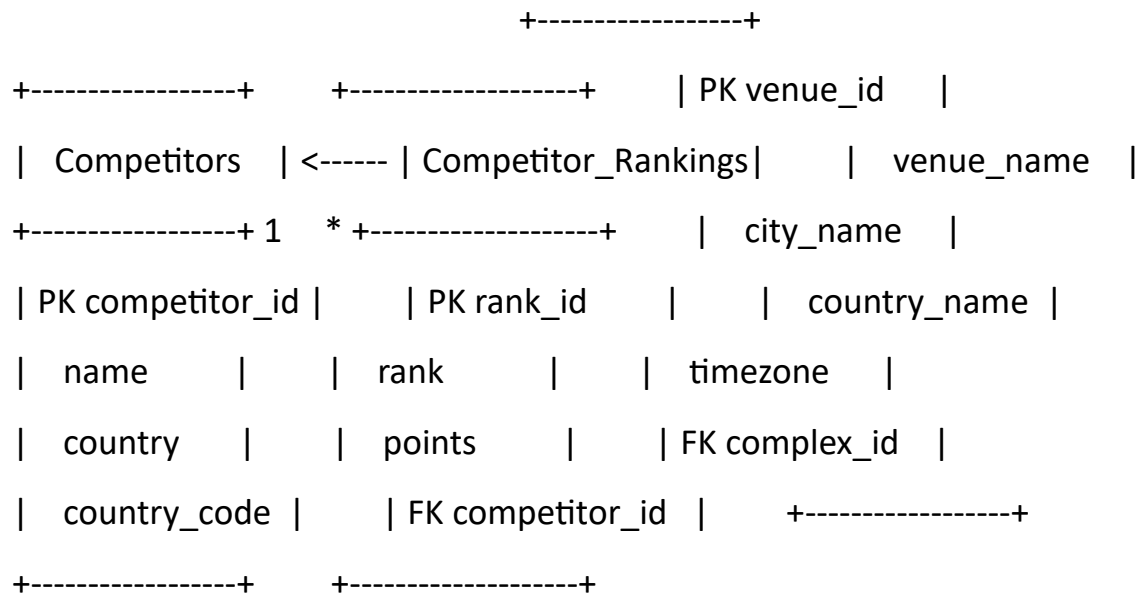


Table Definitions

- **Categories:** Primary storage for competition classifications.
 - *Columns:* category_id (PK, VARCHAR), category_name (VARCHAR)
- **Competitions:** Hierarchical details on tennis tournaments.
 - *Columns:* competition_id (PK, VARCHAR), competition_name (VARCHAR), parent_id (VARCHAR), type (VARCHAR), gender (VARCHAR), category_id (FK, VARCHAR)
- **Complexes:** Parent entities representing physical tournament facilities.
 - *Columns:* complex_id (PK, VARCHAR), complex_name (VARCHAR)
- **Venues:** Specific courts/grounds linked to complexes.
 - *Columns:* venue_id (PK, VARCHAR), venue_name (VARCHAR), city_name (VARCHAR), country_name (VARCHAR), country_code (VARCHAR), timezone (VARCHAR), complex_id (FK, VARCHAR)
- **Competitors:** Player profiles and nationality metadata.
 - *Columns:* competitor_id (PK, VARCHAR), name (VARCHAR), country (VARCHAR), country_code (VARCHAR), abbreviation (VARCHAR)
- **Competitor_Rankings:** Weekly ATP/WTa ranking statistics and points breakdown.

- *Columns:* rank_id (PK, INT IDENTITY), rank (INT), movement (INT), points (INT), competitions_played (INT), competitor_id (FK, VARCHAR)

4. Technical Challenges & Engineering Solutions

Challenge	Root Cause	Engineering Solution
Network Connection Resets (WinError 10054)	ISP-level Deep Packet Inspection (DPI) resetting persistent TCP connections on large JSON payloads.	Configured urllib3 to force IPv4 socket resolution (socket.AF_INET), appended "Connection": "close" HTTP headers, and implemented retry logic with backoff delays.
API Rate Limiting (429 Errors)	Sportradar Trial API enforces a strict throttle of 1 request per second.	Integrated time.sleep(2.5) pauses between API calls and segmented bulk requests into modular python modules.
Referential Integrity Violations	SQL Server rejected bulk insertions when child records were loaded prior to parent entities.	Enforced strict execution order in load_to_sql.py: Primary entities (Categories, Complexes, Competitors) populate fully prior to dependent tables (Competitions, Venues, Competitor_Rankings).

5. Key Analytical Insights & Findings

1. **Tournament Hierarchy:** Over 60% of tracked competitions belong to regional ITF circuits, with doubles tournaments accounting for a major portion of international schedule volume.
2. **Geographic Concentration:** Venue distribution mapping reveals dense clusters in Europe and North America, with major complexes managing up to 12 sub-venues on a single property.

3. **Player Point Distribution:** Competitor point analysis follows a heavy power-law curve — top 5 ranked players hold a disproportionately large share of total points compared to ranks 20–100.